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3D printing non-powered compartments with passive latches

Abstract

A 3D printing apparatus is disclosed herein. The 3D printing apparatus comprises a non-powered compartment defining a chamber to contain build material, and a passive latching element connected to a lateral wall of the compartment to be engaged with a complementary latching element of a platform. A drive mechanism of a 3D printing device is engageable with the platform to apply an upward vertical force to the platform to move the platform upwardly. The complementary latching elements are to: passively latch the platform and the compartment together at a latched position such that upward movement of the platform causes upward movement of the compartment until the compartment is restrained at a sealing position; and to passively unlatch the platform from the compartment, upon further upward movement of the platform.

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Background/Summary

BACKGROUND

(1) Some additive manufacturing or three-dimensional printing systems comprise a removable build unit that interacts with different 3D printing system sub-systems. Some build units comprise a build chamber defining a volume where 3D objects are generated. The build chamber may host a build platform to perform a 3D printing operation in interaction with the 3D printing sub-system in which the build unit resides.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present application may be more fully appreciated in connection with the following detailed description of non-limiting examples taken in conjunction with the accompanying drawings, in which like reference characters refer to like parts throughout and in which:

(2) FIG. **1**A is a schematic diagram showing an example of a 3D printing apparatus with a non-powered compartment and a passive latching element in a first position;

(3) FIG. **1**B is a schematic diagram showing an example of a 3D printing apparatus with a non-powered compartment and a passive latching element in a second position;

(4) FIG. **1**C is a schematic diagram showing an example of a 3D printing apparatus with a non-powered compartment and a passive latching element in a third position;

(5) FIG. **2**A is a schematic diagram showing an example of a passive latching element from a 3D printing apparatus;

(6) FIG. **2**B is a schematic diagram showing an example of a complementary latching element from a moveable platform;

(7) FIG. **3**A is a schematic diagram showing an example of a configuration of a 3D printing apparatus and a platform with the corresponding latching elements;

(8) FIG. **3**B is a schematic diagram showing an example of another configuration of a 3D printing apparatus and a platform with the corresponding latching elements;

(9) FIG. **3**C is a schematic diagram showing an example of another configuration of a 3D printing apparatus and a platform with the corresponding latching elements;

(10) FIG. **3**D is a schematic diagram showing an example of another configuration of a 3D printing apparatus and a platform with the corresponding latching elements;

(11) FIG. **3**E is a schematic diagram showing an example of another configuration of a 3D printing apparatus and a platform with the corresponding latching elements;

(12) FIG. **4**A is a schematic diagram showing an example of another passive latching element from a 3D printing apparatus;

(13) FIG. **4**B is a schematic diagram showing an example of another example of a complementary latching element from a moveable platform;

(14) FIG. **5**A is a schematic diagram showing an example of a configuration of a 3D printing build unit in a 3D printer;

(15) FIG. **5**B is a schematic diagram showing an example of another configuration of a 3D printing build unit in a 3D printer;

(16) FIG. **5**C is a schematic diagram showing an example of another configuration of a 3D printing build unit in a 3D printer; and

(17) FIG. **5**D is a schematic diagram showing an example of another configuration of a 3D printing build unit in a 3D printer.

DETAILED DESCRIPTION

(18) The following description is directed to various examples of additive manufacturing, or three-dimensional printing, apparatus and processes to generate high quality 3D objects. Throughout the present disclosure, the terms “a” and “an” are intended to denote at least one of a particular

element. In addition, as used herein, the term "includes" means includes but not limited to, the term "including" means including but not limited to. The term "based on" means based at least in part on.

(19) For simplicity, it is to be understood that in the present disclosure, elements with the same reference numerals in different figures may be structurally the same and may perform the same functionality.

(20) Some elements in the examples shown herein are drawn in dotted lines to indicate that the elements may be external elements but may interact with the apparatuses or devices being disclosed therein.

(21) 3D printing systems generate a 3D object by executing a series of 3D printing operations. In some 3D printing systems, some of the 3D printing operations are distinct from each other and may be executed by different sub-systems of the 3D printing system (3D printing sub-systems are also referred herein as 3D printing devices). The sub-systems may be different depending on the type of material and 3D printing technology used. Some sub-systems may be physically placed in different locations.

(22) Some removable build units may be attached and detached from the different sub-systems according to the 3D printing system workflow. A build unit may be understood as the module including a build chamber where 3D objects are to be generated during the 3D printing process of a 3D printing system.

(23) Some 3D printing operations may include at least one of loading the removable build unit with build material, heating part of the build unit, selectively solidifying portions of build material from the build unit, ejecting agents (e.g., binding agents, fusing agents, detailing agents, colour agents) to portions of the build material from the build unit, curing the contents of a build unit (e.g., thermally curing), thermally fusing the portions of build material in which fusing agents have been deposited, separating non-solidified build material from the generated 3D objects (i.e., decaking), recycling non-solidified build material, removing 3D objects from the build unit, cleaning the build unit, and the like.

(24) Some sub-systems that perform at least one of the printing operations mentioned above may include at least one of a build material management station, a 3D printer, a curing station, a cleaning station, a deca king station and the like.

(25) Some 3D systems generate 3D objects by selectively processing layers of build material. Suitable powder-based build materials for use in additive manufacturing may include, where appropriate, at least one of polymers, metal powder, and ceramic powder. In some examples build materials may be provided in other forms, such as gels, pastes, and slurries.

(26) As mentioned above, build units may be attached to and detached from different sub-systems in which different 3D printing operations are executed. Typical build units are technically complex independent modules that interact with the different sub-systems of a 3D printing system. Build units comprise a moveable build platform therein to assist in the 3D printing operations execution. Some build units comprise additional mechanisms and equipment to further assist in the 3D printing operations execution. For example, some build units comprise heaters (e.g., resistive heaters or heating blankets) to transfer heat to the contents of the build unit and thereby maintain them at a constant or controlled temperature. Build units also comprise a build platform drive mechanism to cause the movement of the build platform. Hence, build units may comprise expensive and complex equipment that execute 3D printing operations when the build unit is engaged in the appropriate 3D printing sub-system. A fleet of multiple build units raises the cost a 3D printing system since the aforementioned expensive and complex equipment is replicated in each build unit.

(27) A single build unit may be used by different sub-systems to perform different printing operations. Some build unit elements and mechanisms are expensive and are not used in every single sub-system that the build unit interacts with. For this reason, some expensive build unit

mechanisms may be used infrequently throughout a complete build unit use cycle.

(28) In some examples, when a build unit is coupled to a 3D printing sub-system, a sealing operation may be executed. As mentioned above, the contents of the build unit may include powder-based build material which is to be isolated from the elements of the 3D printing sub-system (e.g., engines, electronics, heaters). Handling powder-based build material in the build unit may cause the build material to become airborne and pollute other areas of a 3D printing sub-system if not appropriately sealed, which may damage the elements of the 3D printing sub-system. Therefore, in such examples, a sealing operation may be executed prior the performance of the 3D printing operation of the 3D printing sub-system that the build unit is coupled thereto.

(29) Referring now to the drawings, FIGS. **1**A-**1**C are schematic diagrams showing a cross-sectional side view of an example of a 3D printing apparatus **100** with a non-powered compartment **110** and a passive latching element **115** in different positions. In some examples, the 3D printing apparatus **100** (referred hereinafter as apparatus **100**) is a 3D printing build unit. In other examples, however, the apparatus **100** is a build material reservoir.

(30) The apparatus **100** may be configured as a non-powered single transportable element. The apparatus **100** may be transportable and engageable with different 3D printing sub-systems. In an example, the apparatus **100** is suitable to be used in 3D printing operations in a 3D printer, a build material management station, a decaking station, a curing station, and the like. Therefore, the apparatus **100** may be engageable with receiving interfaces from different 3D printing sub-systems. A more detailed description with reference to a 3D printer is disclosed in some of the examples below (see, e.g., FIGS. **5**A-**5**D).

(31) The apparatus **100** comprises a compartment **110** defining a chamber **112** therein. The compartment **110** is a passive, or non-powered, element by which is meant not comprising any electronic element that needs electrical power to perform its functionality. In some examples, the compartment **110** comprises a lateral wall or a plurality of lateral walls. Additionally, in some examples, the compartment **110** may additionally comprise a top wall. A top wall may be implemented, for example, in the form of a removable sealable lid. The horizontal cross-section of the compartment **110** may be rectangular, circular, rectangular with rounded corners, or any other shape suitable for the generation of a 3D object therein.

(32) In an example in which the apparatus **100** is a build unit, when in use, a 3D printer may generate 3D objects in the chamber **112** out of build material. When it is not in use, the apparatus **100** may not comprise build material or may comprise a full bed of build material and generated 3D objects therein. In another example in which the apparatus **100** is a build material reservoir, when in use, a build material management station may fill the chamber **112** with build material to be used in the generation of a 3D object.

(33) The apparatus **100** is engageable with a platform **120** to perform its functionality. In some examples, the platform **120** is not part of the apparatus **100**. In other examples, the platform **120** is part of the apparatus **100**. The platform **120** is a passive element. The platform **120** comprises a platform body defining an upper surface on which layers of build material can be formed.

(34) In some examples, the platform **120** may be externally controlled to move within the chamber **112** according to the examples of the present disclosure. The platform **120** comprises a platform drive interface **122** engageable with an external drive mechanism (not shown) to cause the platform **120** to move. The drive mechanism is part of an external 3D printing sub-system engageable with the apparatus **100** (e.g., a 3D printer, a curing station, a build material management station). In an example, the external drive mechanism is controllable to apply an upward vertical force to the platform **120** and thereby move the platform **120** vertically upwardly. The external drive mechanism may also move the platform **120** downwardly. In other examples, however, the external drive mechanism may also move the platform laterally or rotate (e.g., tilt) the platform **120** with respect to a horizontal plane.

(35) The apparatus **100** further comprises a passive latching element **115** connected to or integral

with a lateral wall of the compartment **110**. The passive latching element **115** (also referred herein as apparatus latching element **115** or latching element **115**) is not electrically controlled, thereby being to latch and unlatch based on external mechanical interaction. The apparatus latching element **115** is to be engaged with a complementary platform latching element **125** of the platform **120**. In some examples, the platform latching element **125** may be connected to or integral with a lower surface of the platform **120** body to be engaged with the apparatus latching element **115** within the compartment **110**.
(36) In the examples herein, the term “to latch” is used to mean holding two elements together and may not necessarily involve a grip or a holding action. In some examples, the term “latch” may be equivalent to “couple”.
(37) FIG. **1**A illustrates a first position corresponding to a start position, where the platform **120** may be positioned within the compartment **110** at a position below the apparatus latching element **115**. Upon the introduction of the apparatus **100**, the drive mechanism may couple with the platform **120** through the platform interface **122**, so that the height of the platform **120** is controllable through the application of an upward force by the drive mechanism.
(38) The external drive mechanism (not shown) causes the platform **120** to move upwardly until the platform latching element **125** engages with the apparatus latching element **115** (see, e.g., second position illustrated in FIG. **1**B). At this position, the engagement between the platform latching element **125** and the apparatus latching element **115** causes the platform **120** and the compartment **110** to passively latch together at a latched position. In the latched position, an upward movement of the platform **120** causes a corresponding upward movement of the compartment **110**. In the latched position, the drive mechanism may apply an upward vertical force sufficient for lifting the compartment **110**, the platform **120**, and the contents of the compartment **110** (if any) up to a sealing position.
(39) In some examples, the compartment **100** may unlatch from the platform when the compartment is moved into a sealing position and a separate locking mechanism from the 3D printing sub-system restrains the compartment **110** at the sealing position. The locking mechanism may engage with a locking interface from the compartment **110**. The locking mechanism may be a mechanical system actuatable between its locked and unlocked positions. In another example, the external locking mechanism comprises electronic components that enable it to be controlled by an external controller (not shown) to switch between its locked and unlocked positions. The locking element may be implemented as a pin, screw, grip, or any suitable locking mean to secure the compartment **110** (apparatus **100**) to the 3D printing sub-system and restrain the compartment **110**, and the contents of the compartment **110**, at the sealing position.
(40) As it has been illustrated in FIG. **1**C, once the compartment **100** reaches the sealing position, the external drive mechanism may continue to apply an additional upward vertical force to cause the platform **120** to move further upwards and to cause the complementary latching elements **115** and **125** to passively unlatch the platform **120** from the compartment **110**.
(41) Once the compartment **110** is secured at the sealing position and the platform **110** is unlatched from the compartment **110**, the platform **110** may be controlled to move freely within the chamber **112** and execute a 3D printing operation from the 3D printing sub-system (see, e.g., example implementation of FIGS. **3**E and **5**D).
(42) In the example in which the 3D printing sub-system is a 3D printer, the drive mechanism may move the platform **120** to a top part of the compartment **110**. Then, an external layer forming element from the 3D printer (not shown) may spread build material to generate a layer of build material on the platform **120** or on the uppermost build material layer on the platform **120**. Once the layer of build material has been generated, a selective solidification module (not shown) from the 3D printer may selectively solidify portions of the uppermost layer to generate the part of the 3D objects corresponding to the generated layer. Then, the platform **120** may be controlled to move (e.g., downwards) for a distance corresponding to a thickness of the subsequent layer to be

generated. The same printing operations may be executed up to the completion of the 3D objects.
(43) The selective solidification module may selectively solidify portions of the uppermost layer of build material in a number of different ways. In an example, the selective solidification module may selectively solidify portions of a layer of build material in a layer-by-layer basis by depositing printing fluids (e.g., fusing agents, detailing agents, property agents, colour agents). In other examples, the selective solidification module may comprise a laser or a laser array to directly selectively solidify portions of a layer of build material; e.g., Selective Laser Sintering (SLS). In other examples, the selective solidification module may selectively deposit binding agents (e.g., thermally curable binder agents, UV curable binder agents) to a layer of build material in a layer-by-layer basis. In yet other examples, the selective solidification module may use other 3D printing techniques to generate a 3D object, for example, Stereolithography (SLA), Digital Light Processing (DLP), Selective Laser Melting (SLM), or the like.
(44) FIGS. **2**A and **2**B show an example implementation of the complementary latching elements from the examples above. FIG. **2**A shows an example of the apparatus latching element **115** from the 3D printing apparatus **100**. FIG. **2**B shows an example of the platform latching element **125** of the platform **120**.
(45) The apparatus latching element **115** of FIG. **2**A comprises a compressible element **220** in a housing **210** defining a volume therein. The housing **210** is fixed to a lateral wall of the compartment **110**. A first end of the compressible element **220** may be coupled to a lateral wall of the compartment **110** and the second end of the compressible element **220** may be coupled to a follower element **230**. When the compressible element **220** is un-compressed, part of the follower element **230** resides within the housing **210** and part of the follower element **230** is outside of the housing **210**. The housing **210** horizontally constrains the movement of the follower element **230**. Some examples of the compressible element **220** include a spring, a pneumatic element, or any other suitable element that compresses, and un-compresses based on an external force applied.
(46) The follower element **230** is to horizontally move towards the lateral wall of the compartment **110** upon receiving a force from a part of the platform **120** caused by an upward movement of the platform **120**. Upon receiving such force, or upon receiving an increased force, the follower element **230** is to compress the compressible element **220**. The distance that the compressible element **220** is moved corresponds to the horizontal displacement of the follower element **230**. Analogously, upon receiving a reduced force the compressible element **220** expands. When the platform moves upwardly beyond a position at which contact is lost with the follower element **230**, the platform ceases to apply the upward force to the follower element **230**, the follower element **230** the compressible element **220** return to a start configuration.
(47) The housing **210**, the compressible element **220** and the follower element **230** may be integrated as a single latching element **115**. In other words, the latching element **115** is a horizontally constrained resiliently loaded element coupled to a lateral wall of the compartment **110**.
(48) Turning now to FIG. **2**B, the platform latching element **125** comprises a sloped portion **240** coupled to the platform **120**. The sloped portion **240** may be implemented as a protrusion which is sloped with respect the horizontal plane. The follower element **230** of the apparatus latching element **115** and the sloped portion **240** of the platform latching element **125** may be designed in such a way that they engage (e.g., latch) upon surface contact. Therefore, the follower element **230** may include one of a pin, a bearing, a cone and/or any shape suitable to engage with the sloped portion **240** from the platform **120**.
(49) The sloped portion **240** may have a sloped profile in its upper face, so that the sloped upper face engages with the lower face of the follower element **240** upon upward movement of the platform **120** (i.e., engaging from below the compartment latching element **215**). In other examples, the sloped portion may have a first sloped profile in the upper face to engage with the lower face of the follower element **240** upon upward movement of the platform **120**, and a second sloped profile

in the bottom face to engage with the upper face of the follower element **240** upon downward movement of the platform **120** (i.e., engaging from above the compartment latching element **215**). In some additional examples, the first slope profile angle and the second slope profile angle may be the same angle. In other additional examples, the first slope profile angle may be different than the second slope profile angle.

(50) FIGS. **3**A-**3**E depict an example workflow implementation of the complementary latching elements (i.e., apparatus latching element **115** and platform latching element **125**) in a 3D printing operation, for example, a sealing operation.

(51) FIG. **3**A is an example implementation of the apparatus **100** when the apparatus is introduced in a 3D printing sub-system. The apparatus further comprises a stopper **350** in the compartment **110** below the apparatus latching element **210** to inhibit the movement of the platform **120** below the stopper **350**. Once the apparatus **100** is introduced in the 3D printing sub-system, the drive mechanism of the 3D printing sub-system engages with the platform **120** through the platform drive interface **122**.

(52) In FIG. **3**B, the drive mechanism applies an upward force to move the platform **120** upwardly (arrow **320**B). The upward movement of the platform **120**, causes the platform latching element **125** (e.g., the sloped portion **240**) to engage with the apparatus latching element **115** (follower element **230**). This engagement passively couples the platform **120** and the compartment **110** together, so that further upward movement of the platform **120** (arrow **320**B), applied by the drive mechanism, causes an upward movement of the compartment **110** (arrow **310**B). There is no direct contact between the drive mechanism and the compartment **110** and yet the drive mechanism movement causes upward movement of both the platform **120** and the compartment **110**. The platform **120** and the compartment **110** are coupled and move together up to the point that the compartment **110** reaches a sealing position, where it is constrained thereto. In an example, the compartment **110** is constrained (e.g., secured) to the 3D printing sub-system by means of a locking mechanism from the 3D printing sub-system engageable with a locking interface of the compartment **110**.

(53) In FIG. **3**C, the compartment **110** is constrained at the sealing position. Then, upon further upward movement of the platform **120**, the following element **230** may act as a follower mechanism (e.g., cam follower, track follower) with respect to the sloped portion **240** of the platform **120**. In this configuration, the drive mechanism may apply a greater force to the platform **120**, so that the sloped profile **240** applies a corresponding greater force to the follower element **230**. The follower element **230** may then compress the compressible element **220** and thereby move towards the lateral wall of the compartment **110** (i.e., follower mechanism) for a followed distance or displacement. As the follower element **230** acts as a follower mechanism with respect to the sloped portion **240** of the platform **120**, the platform **120** keeps moving upwards (arrow **320**C) for a distance corresponding to the followed distance until the latching elements reach an unlatching position.

(54) FIG. **3**D shows the position in which the platform latching element **125** is about to unlatch from the apparatus latching element **115**. At this point, the drive mechanism has to apply a force to the platform **120** greater than a force threshold to cause a further displacement of the follower element **230** such that the platform **120** loses the contact point with the follower element **230** and thereby unlatch the platform **120** from the compartment **110**.

(55) FIG. **3**E shows the unlatched configuration of the platform **120** and the compartment **110** at the sealing position. In this configuration, the drive mechanism freely moves the platform **120** upwardly (arrow **320**E) to allow a 3D printing operation to be executed in the 3D printing sub-system in which the compartment **110** is inserted thereto. As mentioned above, in an example in which the 3D printing sub-system is a 3D printer, the drive mechanism may move the platform **120** to a top part of the compartment **110** and start forming build material layers thereon to be subsequently selectively solidified to generate a 3D object.

(56) FIGS. **4**A and **4**B show an alternative example implementation of the complementary latching elements from the examples above. FIG. **4**A shows an example of the apparatus latching element **115** from the 3D printing apparatus **100**. FIG. **4**B shows an example of the platform latching element **125** of the platform **120**.
(57) The apparatus latching element **115** is the same as or similar to the platform latching element **125** of FIG. **2**B, thereby comprising a sloped portion **240** coupled to the lateral wall of the compartment **110**. The platform latching element **125** is the same as or similar to the apparatus latching element **115** of FIG. **2**A, thereby being a horizontally constrained resiliently loaded element coupled to the platform **120**. In this example, the functionality of the interaction between apparatus latching element **115** and the platform latching element **125** may be the same as or similar to as respectively the functionality of the interaction between the platform latching element **125** and apparatus latching element **115** of FIGS. **2**A-B and **3**A-E.
(58) Therefore, upon upward movement of the platform **120**, the latching element of the apparatus **115** may engage with the complementary platform latching element **125** to passively couple the platform **120** and the compartment **110** together, so that upward movement of the platform **120** causes upward movement of the compartment **110** until the compartment **110** is restrained at the sealing position. Upon further upward movement of the platform **120**, the platform complementary latching element **125** acts as a follower mechanism with respect to the sloped portion **240** to move the complementary platform latching element **125** away from the lateral wall of the compartment **110** until the latching elements reach an unlatching position.
(59) FIGS. **5**A-D depict an example workflow implementation of a sealing operation of a 3D printing apparatus (e.g., apparatus **100**) within a 3D printing sub-system **500**. Some elements and processes of FIGS. **5**A-D may correspond to some elements of FIGS. **1**A-C, **2**A-B, and **4**A-B and some processes of FIGS. **3**A-E. In some examples, the apparatus **100** is a build unit and the 3D printing sub-system **500** is a 3D printer.
(60) FIG. **5**A is an example configuration of the apparatus **100** when the apparatus is introduced into a 3D printing sub-system **500** (e.g., a 3D printer), also referred to as 3D printing device **500**. The 3D printing device **500** comprises an enclosure **510** to host the apparatus **100**, the driving mechanism **520** to apply a vertical force to the platform **120** and move the platform **120** upwardly, a sealing element **550** to seal the contents of the apparatus **100** and a locking mechanism (not shown). Once the apparatus **100** is introduced into the enclosure **510**, the drive mechanism **520** engages with the platform **120** through the platform drive interface **122**. In FIG. **5**A, the relative position between the apparatus **100** and the platform **120** may be the same as or similar to the relative position of the apparatus **100** and the platform **120** depicted in FIG. **3**A.
(61) In FIG. **5**B, the drive mechanism **520** applies an upward force to move the platform **120** upwardly. The upward movement of the platform **120** causes the platform latching element **125** to engage with the apparatus latching element **115**. This engagement passively couples the platform **120** and the compartment **110** together, so that an upward movement of the platform **120** causes an upward movement of the compartment **110**. Therefore, the drive mechanism **520** causes the movement of both the platform **120** and the compartment **110** without being in direct contact with the compartment **110**. The platform **120** and the compartment **110** are coupled and move together up to the point that the compartment **110** reaches a sealing position. In FIG. **5**B, the relative position between the apparatus **100** and the platform **120** may be the same as or similar to the relative position of the apparatus **100** and the platform **120** depicted in FIG. **3**B.
(62) In an example, the apparatus latching element **115** and the platform latching element **120** may be the same as the corresponding complementary latching elements of FIGS. **2**A and **2**B.
(63) In another example, the apparatus latching element **115** and the platform latching element **120** may be the same as the corresponding complementary latching elements of FIGS. **4**A and **4**B. In yet other example, the apparatus latching element **115** and the platform latching element **120** may be any other equivalent complementary latching mechanism suitable for the examples disclosed

herein.

(64) In FIG. **5**C, the compartment **110** reaches the sealing position, and a top portion of the compartment makes contact with a sealing element **550**. The sealing element may be any suitable element such that at the sealing position, inhibits the contents within the chamber **112** from escaping the chamber **112**. In some examples, the sealing element **550** may be a compressible foam of a geometrical shape suitable for isolating the contents of the chamber **112** (e.g., circular-shaped, rectangular-shaped, torus-shaped). In other examples, the sealing element **550** may be a compressible rubber. In yet other examples, the sealing element **550** may not be compressible.

(65) Once the compartment **110** reaches the sealing position, the locking element **570** secures the apparatus **100** to the 3D printing device **500** and restrain the apparatus at the sealing position, so that there is no relative movement between the compartment **110** and the enclosure **510**. In FIG. **5**C, the relative position between the apparatus **100** and the platform **120** may be the same as or similar to the relative position of the apparatus **100** and the platform **120** depicted in FIGS. **3**C and **3**D.

(66) In FIG. **5**D, once the compartment **110** is restrained at the sealing position, the drive mechanism **520** is to apply an upward force to the platform **120** greater than a force threshold to cause platform latching element **125** to unlatch from the apparatus latching element **115** (see, e.g., example of FIG. **3**D). Once the complementary latching elements **115** and **125** unlatch, the drive mechanism **520** freely moves the platform **120** upwardly according to the 3D printing operation to be executed in the 3D printing device **500** (see, e.g., example of FIG. **3**E). As mentioned above, in an example in which the 3D printing device **500** is a 3D printer, the drive mechanism **520** may move the platform **120** to a top part of the compartment **110** and start building build material layers thereon to be subsequently selectively solidified to generate a 3D object.

(67) As used herein, the terms “substantially” and “about” are used to provide flexibility to a range endpoint by providing a degree of flexibility. The degree of flexibility of this term can be dictated by the particular variable and would be within the knowledge of those skilled in the art to determine based on experience and the associated description herein.

(68) The drawings in the examples of the present disclosure are some examples. It should be noted that some units and functions of the procedure may be combined into one unit or further divided into multiple sub-units. What has been described and illustrated herein is an example of the disclosure along with some of its variations. The terms, descriptions and figures used herein are set forth by way of illustration. Many variations are possible within the scope of the disclosure, which is intended to be defined by the following claims and their equivalents.

(69) There have been described example implementations with the following sets of features:

(70) Feature set 1: A 3D printing apparatus comprising: a non-powered compartment defining a chamber to contain build material; and a passive latching element connected to a lateral wall of the compartment to be engaged with a complementary latching element of a platform; wherein a drive mechanism of a 3D printing device engageable with the platform is to apply an upward vertical force to the platform to move the platform upwardly; and wherein the complementary latching elements are to: passively latch the platform and the compartment together at a latched position such that upward movement of the platform causes upward movement of the compartment until the compartment is restrained at a sealing position; and passively unlatch the platform from the compartment, upon further upward movement of the platform.

(71) Feature set 2: A 3D printing apparatus with feature set 1, wherein the latching element of the apparatus is a horizontally constrained resiliently loaded element coupled to a lateral wall of the compartment that is to: (i) upon upward movement of the platform, engage with a sloped portion of the complementary latching element of the platform to passively couple the platform and the compartment together, so that upward movement of the platform causes upward movement of the compartment until the compartment is restrained at the sealing position; and (ii) upon further upward movement of the platform, act as a follower mechanism with respect to the sloped portion

to move the latching element towards the lateral wall of the compartment until the latching elements reach an unlatching position.

(72) Feature set 3: A 3D printing apparatus with any preceding feature set 1 to 2, wherein the horizontally constrained resiliently loaded element comprises at least one of a spring, a pneumatic element, a pin, a bearing and/or a cone.

(73) Feature set 4: A 3D printing apparatus with any preceding feature set 1 to 3, wherein the latching element of the apparatus comprises a sloped portion coupled to a lateral wall of the compartment to: (i) upon upward movement of the platform, engage with the platform complementary latching element, which is a horizontally constrained resiliently loaded element, to passively couple the platform and the compartment together, so that upward movement of the platform causes upward movement of the compartment until the compartment is restrained at the sealing position; and (ii) upon further upward movement of the platform, cause the platform complementary latching element to act as a follower mechanism with respect to the sloped portion to move the complementary latching element from the platform away from the lateral wall of the compartment until the latching elements reach an unlatching position.

(74) Feature set 5: A 3D printing apparatus with any preceding feature set 1 to 4, further comprising a stopper in the compartment below the apparatus latching element to inhibit a movement of the platform below the stopper.

(75) Feature set 6: A 3D printing apparatus with any preceding feature set 1 to 5, further comprising a locking interface engageable with a locking mechanism from a 3D printing device to secure the 3D printing apparatus to the 3D printing device and restrain the apparatus at the sealing position.

(76) Feature set 7: A 3D printing apparatus with any preceding feature set 1 to 5, wherein the 3D printing device is a 3D printer. a 3D printing apparatus receiving interface to receive an external 3D printing apparatus that comprises:

(77) Feature set 8: A passive platform comprising: a platform body defining an upper surface on which layers of build material can be formed; a drive interface engageable with a drive mechanism from a 3D printing device to move the platform vertically; a passive platform latching element connected to a lower surface of the platform body to be engaged with a complementary latching element of a compartment, wherein the complementary latching elements are to: passively latch the platform and the compartment together at a latched position such that upward movement of the platform causes upward movement of the compartment until the compartment is restrained at a sealing position, and passively unlatch the platform from the compartment, upon further upward movement of the platform.

(78) Feature set 9: A passive platform with feature set 8, wherein the latching element of the platform is a horizontally constrained resiliently loaded element coupled to the platform that is to: (i) upon upward movement of the platform, engage with a sloped portion of the complementary latching element of the compartment to passively couple the platform and the compartment together, so that upward movement of the platform causes upward movement of the compartment until the compartment is restrained at the sealing position; and (ii) upon further upward movement of the platform, act as a follower mechanism with respect to the sloped portion to move the latching element away from the lateral wall of the compartment until the latching elements reach an unlatching position.

(79) Feature set 10: A passive platform with any preceding feature set 8 to 9, wherein the horizontally constrained resiliently loaded element comprises at least one of a spring, a pneumatic element, a pin, a bearing and/or a cone.

(80) Feature set 11: A passive platform with any preceding feature set 8 to 10 wherein the latching element of the platform comprises a sloped portion coupled to the platform to: (i) upon upward movement of the platform, engage with the compartment complementary latching element, which is a horizontally constrained resilient loaded element, to passively couple the platform and the compartment together, so that upward movement of the platform causes upward movement of the

compartment until the compartment is restrained at the sealing position; and (ii) upon further upward movement of the platform, cause the compartment complementary latching element to act as a follower mechanism with respect to the sloped portion to move the complementary latching element from the compartment towards the lateral wall of the compartment until the latching elements reach an unlatching position.
(81) Feature set 12: A passive platform with any preceding feature set 8 to 11, wherein the sloped profile comprises: (i) a first upper slope to engage with the compartment latching element when the platform engages from below the compartment latching element; and (ii) a second lower slope to engage with the compartment latching element when the platform engages from above the compartment latching element.
(82) Feature set 13: A hosting device with any preceding feature set 8 or 12, wherein the 3D printing device is a 3D printer.
(83) Feature set 14: A 3D printing build unit comprising: a non-powered compartment defining a chamber to contain build material; and a passive latching element connected to a lateral wall of the compartment to be engaged with a complementary latching element of a platform, the passive latching element being a horizontally constrained resiliently loaded element, wherein a drive mechanism of a 3D printing device engageable with the platform is to apply an upward vertical force to the platform to move the platform upwardly; and wherein the complementary latching elements are to: upon upward movement of the platform, engage with a sloped portion of the complementary latching element of the platform to passively couple the platform and the compartment together, so that upward movement of the platform causes upward movement of the compartment until the compartment is restrained at a sealing position; and
upon further upward movement of the platform, act as a follower mechanism with respect to the sloped portion to move the latching element towards the lateral wall of the compartment until the latching elements reach an unlatching position.
(84) Feature set 15: A 3D printing build unit with feature set 14, further comprising a locking interface engageable with a locking mechanism from a 3D printer to secure the build unit to the 3D printer and restrain the build unit at the sealing position.

Claims

1. A 3D printing device comprising: a compartment defining a chamber to contain build material, the compartment comprising a passive latching element; a platform comprising a complementary passive latching element; and a drive mechanism configured to engage with the platform; wherein the passive latching element is connected to a lateral wall of the compartment, wherein the passive latching element is configured to engage with the complementary passive latching element of the platform, wherein one of the passive latching element or the complementary passive latching element comprises a compressible element; wherein the drive mechanism is configured to apply an upward vertical force to the platform to move the platform upwardly to a latching position, then to a sealing position, and then to an unlatching position; wherein the passive latching element and the complementary passive latching element are configured to passively latch the platform and the compartment together at the latching position; wherein an upward movement of the platform from the latching position to the sealing position causes an upward movement of the compartment until the compartment is restrained at the sealing position; wherein the passive latching element and the complementary passive latching element are configured to passively unlatch the platform from the compartment upon further upward movement of the platform from the sealing position to the unlatching position.
2. The 3D printing device of claim 1, wherein the passive latching element of the compartment comprises the compressible element and is a horizontally constrained resiliently loaded element coupled to the lateral wall of the compartment that is configured to: upon the upward movement of

the platform, engage with a sloped portion of the complementary passive latching element of the platform to passively couple the platform and the compartment together, so that the upward movement of the platform causes the upward movement of the compartment until the compartment is restrained at the sealing position; and upon the further upward movement of the platform, act as a follower mechanism with respect to the sloped portion to move the passive latching element towards the lateral wall of the compartment until the passive latching element and the complementary passive latching element reach an unlatching position.

3. The 3D printing device of claim 2, wherein the horizontally constrained resiliently loaded element comprises at least one of a pin, a bearing and/or a cone, and wherein the compressible element comprises a spring or a pneumatic element.

4. The 3D printing device of claim 1, wherein the complementary passive latching element comprises the compressible element and is a horizontally constrained resiliently loaded element coupled to the platform, and wherein the passive latching element of the compartment comprises a sloped portion coupled to the lateral wall of the compartment configured to: upon the upward movement of the platform, engage with the complementary passive latching element to passively couple the platform and the compartment together, so that the upward movement of the platform causes the upward movement of the compartment until the compartment is restrained at the sealing position; and upon the further upward movement of the platform, cause the complementary passive latching element to act as a follower mechanism with respect to the sloped portion to move the complementary passive latching element from the platform away from the lateral wall of the compartment until the passive latching element and the complementary passive latching element reach an unlatching position.

5. The 3D printing device of claim 1, further comprising a stopper in the compartment below the passive latching element configured to inhibit a movement of the platform below the stopper.

6. The 3D printing device of claim 1, further comprising a locking interface and a locking mechanism configured to secure the compartment to the 3D printing device and restrain the compartment at the sealing position.

7. The 3D printing device of claim 1, wherein the 3D printing device is a 3D printer.

8. A 3D printing device comprising: a compartment defining a chamber to contain build material comprising a passive latching element; a platform comprising a complementary passive latching element; and a drive mechanism; wherein the passive latching element is connected to a lateral wall of the compartment and is configured to be engaged with the complementary passive latching element of the platform, the passive latching element being a horizontally constrained resiliently loaded element; wherein the drive mechanism is engageable with the platform and is configured to apply an upward vertical force to the platform to move the platform upwardly to a latching position, then to a sealing position, and then to an unlatching position; wherein the passive latching element is configured to engage with a sloped portion of the complementary passive latching element of the platform to passively couple the platform and the compartment together at the latching position; wherein an upward movement of the platform from the latching position to the sealing position causes an upward movement of the compartment until the compartment is restrained at the sealing position; and wherein upon a further upward movement of the platform from the sealing position to the unlatching position, the passive latching element is configured to function as a follower mechanism with respect to the sloped portion to move the passive latching element towards the lateral wall of the compartment until the passive latching element and the complementary passive latching element reach an unlatching position.

9. The 3D printing device of claim 8, further comprising a locking interface and a locking mechanism configured to secure the compartment to the 3D printing device and restrain the compartment at the sealing position.